Depreciation of Organizational Knowledge in Industrial Production (Argote, Beckman & Epple 1990)

Empirical econometric analysis of manufacturing production curves shows organizational knowledge depreciates rapidly over interruptions, proving that institutional memory behaves as a perishable stock rather than a permanent asset.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL, ORGANIZATIONAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Traditional industrial learning curves assume organizational knowledge accumulates monotonically with cumulative output (the classic Wright learning model). Argote, Beckman, and Epple (1990) demonstrated that organizational knowledge depreciates rapidly over time: when production ceases or rates decline, unit labor costs spike upon resumption, proving that institutional knowledge is a decaying capital stock requiring ongoing operational throughput to maintain.

EVIDENCE AND CAUSAL ASSESSMENT

Argote et al. estimated learning curve models incorporating depreciation parameters (\delta) alongside cumulative experience (K = λ K_{t-1} + Q):

- Models assuming zero depreciation (λ = 1) produced biased overestimates of cumulative productivity gains and failed to forecast cost spikes following production hiatuses.

- In plants experiencing production halts or layoffs, unit labor hours required to assemble successive units reverted toward baseline levels within months, reflecting loss of tacit coordination, worker attrition, and assembly routine drift.

EFFECT SIZE AND PERSISTENCE

The annual depreciation rate of organizational knowledge in high-complexity assembly operations was estimated between 15\% and 25\% per month during periods of inactivity. An interruption of just three months erased up to 40\% of the productivity gains accumulated over the previous year of continuous production.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as robust (<20%). Controlled for tooling changes, engineering design revisions, worker strikes, and material shortages. Variance in depreciation rates across plants correlated with the degree to which production knowledge was codified in specialized tooling and software versus embodied exclusively in informal team tacit routines.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Knowledge depreciation does not mean individual workers develop neurological amnesia; rather, the collective organizational architecture (communication handoffs, informal coordination channels, assembly rhythms) desynchronizes when not actively exercised.

WHAT ACTION IS REASONABLY JUSTIFIED

Operational planning and financial models must discount cumulative experience with an explicit decay factor when production schedules or project staffing face planned hiatuses, and allocate explicit re-ramping and routine refresher budgets.

VERIFIED SOURCES & DOI IDENTIFIERS

- Argote, L., Beckman, S. L., & Epple, D. (1990). The Persistence and Transfer of Learning in Industrial Settings. Management Science, 36(2), 140–154. https://doi.org/10.1287/mnsc.36.2.140

- de Holan, P. M., & Phillips, N. (2004). Remembrance of Things Past? The Dynamics of Organizational Forgetting. Management Science, 50(11), 1603–1613. https://doi.org/10.1287/mnsc.1040.0273